# When Do Compression Errors Cooperate? Exact-Byte Hessian Geometry for Composite Language-Model Compression

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## Abstract

Low-bit quantization, pruning, scale protection, and low-rank correction can each be benign for small perturbations, suggesting that distributing compression across mechanisms may avoid pushing any one beyond its local *comfort zone*. Testing this hypothesis is easy to bias because heterogeneous codecs carry different indices, factors, scales, descriptors, and alignment overhead. We formulate composite post-training compression as a discrete exact-serialized-byte problem and decompose its local loss into gradient, self, cross, proxy-mismatch, and path-curvature terms in a declared positive-semidefinite geometry. Our audited Pythia-70M probe shows why orthogonality alone is insufficient: sparse and low-rank repairs are nearly orthogonal, yet a conservative component-scaled Q+S+L candidate loses to Q+L by +0.558 perplexity when tail-padded to the same 3,248,832-byte final file; its natural encoding leaves 15,680 bytes unused. Optimal Brain Surgeon (OBS) re-estimation meanwhile improves a fixed sparse payload at zero additional bytes. Three separate seed-17 scalability-smoke jobs over Pythia-70M, OPT-125M, and Qwen3-0.6B repeat the near-orthogonal S–L geometry and the positive strict-QSL-minus-QL endpoint sign for similarly underfilled candidates, without being pooled or promoted to multi-seed evidence. We release a round-trip research codec, loss-landscape probes, a fail-closed scale runner, and a training-dependence taxonomy separating frozen PTQ from backward-assisted and fine-tuned methods.

## 1. Introduction

Weight-only post-training compression is no longer a single-knob problem. Second-order quantizers, activation-aware scales, one-shot pruning, sparse outliers, and low-rank error reconstruction all preserve different parts of a pretrained model (Frantar et al., 2023; Frantar & Alistarh, 2023; Lin et al., 2024; Dettmers et al., 2024; Zhang et al., 2024). This diversity motivates a natural hypothesis: if every mechanism has a small-perturbation regime in which loss grows slowly, then splitting a severe compression target across several mechanisms could remain inside several comfort zones and outperform concentrating all repair bytes in one mechanism over the same mandatory quantized base.

Joint mechanisms and their interactions are not new by themselves. SLiM combines quantization, semi-structured sparsity, and low-rank compensation; Harma et al. analyze Q/S order and non-orthogonality; OBR applies Hessian compensation to joint quantization and sparsification; and prior rate–distortion work couples Hessian or Fisher sensitivity to entropy or bit allocation (Mozaffari et al., 2025; Harma et al., 2025; Guo et al., 2026; Choi et al., 2017; Orr et al., 2025; Young, 2025). We therefore study a narrower reproducibility question: what conclusions survive when realized heterogeneous endpoints are decoded from a declared self-describing artifact and compared by natural and final file bytes?

Two gaps obstruct a clean test. First, a nominal bit rate does not determine a heterogeneous artifact's size. Sparse corrections need support indices and row pointers; low-rank corrections need two factors; scales and codebooks need descriptors; containers need headers and alignment. A value-stream calculation can therefore call two endpoints equal-rate even when their decoded files differ. Second, a small Hessian-weighted perturbation is not itself an accuracy certificate. The task Hessian may be indefinite, a PSD proxy may not match it, the first-order term need not vanish, and curvature can drift along the path to the final codec endpoint.

We study these gaps through *exact-byte Hessian diagnostics*. A deterministic research codec serializes the selected tensors, round-trips every endpoint, and charges codes,

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scales, sparse state, factors, descriptors, and padding. In a declared PSD activation-Gram geometry, realized Q, S, and L increments are decomposed into self and signed cross terms. A 13-point interpolation then checks whether the local proxy tracks held-out NLL all the way to the only deployable point, $\epsilon = 1$.

Our contributions are:

- an executable, byte-faithful protocol and fail-closed artifact validator, exact under the declared research codec, for quantization (Q), scale protection, sparse residuals (S), fixed-support Optimal Brain Surgeon (OBS) value repair, low-rank residuals (L), and their compositions;
- a theory that states what PSD near-orthogonality does guarantee and a stricter same-byte exchange condition that includes gradient, interactions, path error, and discrete byte overhead;
- a parameter-utilization analysis based on held-out loss recovered per added file byte, with zero-byte repairs reported separately; and
- an evidence-aware comparison framework separating data-free PTQ, calibrated frozen PTQ, backward-assisted correction, and global fine-tuning/quantization-aware training (QAT).

The current strict result is intentionally negative. On one Pythia-70M run over 6 selected MLP tensors, Q+L and scaled Q+S+L are both 3,248,832 bytes, but their perplexity increases are 6.762 and 7.320. This verified physical endpoint shows that a logical payload estimate is insufficient for the operational comparison. Its strict QSL natural artifact is 15,680 bytes below Q+L and is tail-padded to equal final length, so this result rejects the tested conservative per-layer candidate rather than a budget-exhausted global frontier. It does not establish that composite compression is universally inferior. We additionally execute all native endpoints in three separate single-seed scalability-smoke jobs: full Pythia-70M MLP scope and depth-stratified OPT-125M and Qwen3-0.6B MLP scopes. All three retain a positive strict-scaled-QSL-minus-QL NLL sign despite near-orthogonal S–L proxy geometry. This repeated diagnostic enlarges the execution scope, but unequal tensor scopes and one seed preclude a pooled or model-scale conclusion.

## 2. Related Work and Comparison Lanes

**Frozen post-training methods.** GPTQ uses second-order information to update remaining weights during quantization, while SparseGPT applies an analogous one-shot view to pruning (Frantar et al., 2023; Frantar & Alistarh, 2023). Wanda replaces the second-order solve with an activation–magnitude score (Sun et al., 2024). AWQ searches activation-aware scales that protect salient channels (Lin et al., 2024). SpQR and SqueezeLLM combine dense low-bit state with sparse high-precision state, but the latter's Fisher-sensitivity path requires loss gradients and therefore belongs in a different compute lane (Dettmers et al., 2024; Kim et al., 2024). LQER, QERA, EoRA, and SLiM reconstruct compression error in low-rank or joint sparse/quantized representations (Zhang et al., 2024; 2025; Liu et al., 2024; Mozaffari et al., 2025). OBR applies closed-form Hessian compensation to joint quantization and sparsification (Guo et al., 2026). Optimal Brain Surgeon and Optimal Brain Compression already establish the broader second-order correction lineage (Hassibi & Stork, 1992; Frantar & Alistarh, 2022). Fixed-support OBS refitting, low-rank reconstruction, and Q/S/L composition are therefore baselines and mechanisms in this paper, not base novelty claims.

**Interaction, orthogonality, and rate precedents.** Harma et al. prove that magnitude sparsification and max-scaled quantization are generally non-orthogonal and analyze their order-dependent compound error (Harma et al., 2025). Their transformation-level notion differs from our signed, order-specific cross term under a common local PSD proxy, but it precludes any claim that this paper is the first interaction or orthogonality analysis. ProjQ likewise coordinates quantization noise with a low-rank adapter through orthogonal subspace projection (Yu et al., 2026). Choi et al. combine Hessian-weighted distortion with Huffman coding and entropy-constrained scalar quantization (Choi et al., 2017); Orr et al. study variable-length weight formats and Fisher rate allocation (Orr et al., 2025); and Radio optimizes an LLM rate–distortion objective (Young, 2025). Our narrower target is an executable, byte-faithful audit under a declared research codec: every compared endpoint is round-trip decoded, its natural and padded file bytes are separated, and its signed local-PSD interactions are checked against realized NLL. This is not a claim of the first Hessian-aware rate allocation or the first compressed bitstream.

**End-loss, adaptive, and trained recovery.** GuidedQuant uses end-loss information and YAQA performs full-model adaptive rounding; these works make the evaluation objective more explicit but use different optimization budgets (Kim et al., 2025; Tseng et al., 2026). Q-Palette targets fractional-bit allocation with codebooks (Lee & Song, 2025). LiftQuant adds block correction and an optional end-to-end stage, while FPTQuant learns function-preserving transformations (He et al., 2026; van Breugel et al., 2026). SliderQuant learns fused channel scales and small low-rank auxiliary repairs under local sliding-window reconstruction, making it a close lane-B

*Table 1.* Training-dependence lanes used throughout the comparison.

| Lane | Backward? | Intervention |
|---|---|---|
| A0 | No | Data-free frozen compression |
| A1 | No | Calibration-only frozen compression |
| B | Yes | Encoder-side gradient/Hessian–vector products (HVPs), no global recovery |
| C | Yes | Learned or globally fine-tuned recovery |

analogue to our scale-plus-low-rank parameter-utilization question (Wang et al., 2026). Recent low-rank residual methods further refine where error should be preserved or reconstructed (Cho et al., 2026; Park et al., 2026; Li et al., 2026). We keep data-free frozen, calibrated no-backward, backward-assisted, and globally trained methods in separate lanes because these are different experimental interventions.

The selected primary-source protocol and payload audit is reported in Table 5. It is a literature taxonomy, not a protocol-matched empirical leaderboard.

## 3. Exact-Byte Composite Compression

**Scope and objective.** Let $\mathcal{T}$ be an ordered manifest of eligible weight matrices and $w_0$ their dense FP16 values. A codec endpoint $z$ decodes to $\widehat{w}(z)$ and has on-disk length $\mathcal{B}(z)$. For a fixed manifest, model, and held-out distribution, the discrete problem is

$$\min_{z\in\mathcal{Z}} \mathcal{L}(\widehat{w}(z)) \quad \text{s.t.} \quad \mathcal{B}(z) \le B. \tag{1}$$

The same $\mathcal{T}$ is used for every comparator; all other model weights remain dense. Thus bytes in this paper describe the selected-tensor artifact, not a whole checkpoint or inference package.

**Byte-faithful codec.** For each tensor, the artifact stores packed signed Q codes, FP16 row or row–column-block scales, optional FP16 sparse values with fixed-width compressed sparse row (CSR) support, and optional FP16 low-rank factors. A deterministic manifest stores shapes, dtypes, stream offsets, and alignment. We report both natural bytes and final file bytes after 64-byte alignment and tail padding, then verify file length, SHA-256, decoding, and element-wise equality with the evaluated FP16 endpoint. This is an auditable research format, not a claim of production kernel support.

**Candidate mechanisms.** Q is symmetric row-wise 4-bit round-to-nearest quantization. Block-scale protection solves bounded row–column-block multipliers in the local PSD metric and stores the resulting FP16 scales. S selects a residual support by Wanda-style activation–magnitude importance; a matched control stores the naive residual values. OBS then re-estimates values on the *same* support, replacing already-paid values without changing the artifact layout. L applies a covariance-whitened SVD to the post-Q residual and stores two FP16 factors. Component scales for Q/S/L are solved jointly and folded into their existing FP16 state, so a zero-byte claim is accepted only after natural and file-byte verification. The verified legacy endpoints use the declared Q→S→L construction order; the new unexecuted matrix also exposes L→S as an order ablation. We call Q+L a *single repair over the mandatory Q base*, and Q+S+L a composite repair; neither phrase means that Q is absent.

**Implemented PSD proxy.** For layer $j$, calibration activations define a covariance $C_j$ and

$$q_j(\Delta W) = \tfrac{1}{2}\,\mathrm{tr}(\Delta W C_j \Delta W^\top),$$
$$\rho_H(a,b) = \frac{\sum_j \langle a_j, b_j\rangle_{C_j}}{\sqrt{\sum_j \|a_j\|_{C_j}^2 \sum_j \|b_j\|_{C_j}^2}}. \tag{2}$$

The reported normalized cost is $\sum_j \|\Delta W_j\|_{C_j}^2 / \sum_j \|W_j\|_{C_j}^2$. Each empirical Gram receives a $10^{-5}$ mean-diagonal ridge. Matrices with relative negative spectrum worse than $10^{-7}$ fail; accepted float32 numerical error is repaired once, and an eight-machine-epsilon target and the applied diagonal shift are recorded per layer. Damping and the repair are identity shifts, so the final spectral lower bound is derived from one validated eigenspectrum instead of repeating cubic audit decompositions. Whitened fitting retains one separate cached eigendecomposition used to construct its square-root maps. The same prepared $C_j$ is used by selection, OBS, endpoint costs, Gram terms, and $\rho_H$. Whitened SVD has a separately disclosed numerical fit metric $\widetilde{C}_j$: eigenvalues below a configurable fraction of $\lambda_{\max}(C_j)$ are floored before forming the square root and inverse square root. The runner records that fraction, the number of clipped eigenvalues, and the resulting condition number per layer. Every decoded candidate is still scored in $C_j$; factorizer regularization is not silently presented as the common scoring metric. For the unexecuted large-model smoke, each residual fit uses a seeded randomized SVD with four oversampling vectors and two power iterations; the resolved solver is recorded per layer. Full SVD remains an explicit small-matrix control, so computational approximation is not hidden inside the geometry ablation.

**Discrete strict-rate selection.** For each layer and target, we enumerate feasible low-rank ranks. At each rank, the largest CSR support fitting the requested serialized cap is constructed; the candidate with minimum activation-Gram cost is retained. The strict QSL control uses that layer's Q+L artifact as its cap, reserves alignment slack, and is rejected before NLL evaluation if the aggregate serialized file exceeds the Q+L file. This is a deterministic layerwise constrained search, not a claim of global discrete optimality. It yields an executable endpoint rather than a fractional rank or value-only relaxation. A relaxed QSL point is reported separately to show the effect of the next discrete allocation step. The new, unexecuted aggregate allocator removes the forced-composition condition: each layer may select Q, Q+$S_{\mathrm{OBS}}$, Q+L, or an enumerated QSL state. It adds intermediate supports and 1.25/1.5/2.0 multiples of the local Q+L repair allowance beyond base Q near the strongest rank states, thereby permitting enumerated cross-layer byte borrowing. It prunes an additive one-layer byte/cost surrogate and checks every surviving combination with the exact multi-layer serializer. The additive pruning is heuristic; feasibility is exact under the research codec, while optimality remains limited to this enumerated budget-band state set. We therefore do not call it a globally optimal frontier. Component scaling is fitted only after the unscaled QSL allocation is selected, so the scaled row is a fixed-allocation post-selection ablation rather than the optimum scaled candidate over the whole frontier.

**Endpoint gate.** For every retained endpoint, we evaluate $w(\epsilon) = w_0 + \epsilon(\widehat{w} - w_0)$ on held-out text at 13 fixed values from 0 to 1. The first five points through $\epsilon = 1/8$ fit a local linear–quadratic model; the full path measures proxy/NLL correlation. Only $\epsilon = 1$ is serialized and eligible for a rate–accuracy conclusion.

## 4. Theory and Diagnostic Predictions

**Curvature is not automatically a geometry.** Let $\mathcal{L}(w)$ be held-out loss, $g = \nabla\mathcal{L}(w_0)$, and $J_0 = \nabla^2\mathcal{L}(w_0)$. The task Hessian $J_0$ may be indefinite, so $x^\top J_0 x$ is not a squared norm and a "Hessian cosine" based on it need not be an angle. We reserve $H \succeq 0$ for a declared positive-semidefinite metric: an empirical Fisher, a generalized Gauss–Newton matrix, the activation Gram proxy used in our experiments, or $J_0 + \lambda I$ only when $\lambda$ is large enough to make it PSD. We write $\langle a, b\rangle_H = a^\top H b$ and $\|a\|_H^2 = a^\top H a$; zero-energy directions are excluded when a correlation is reported.

**Self, cross, and path terms.** For sequential compression stages, define the realized increments $\delta_i = w_i - w_{i-1}$ and $\delta = \sum_i \delta_i$. This telescoping identity is exact, although each $\delta_i$ depends on stage order. A local expansion around $w_0$ can be written

$$\begin{aligned} \mathcal{L}(w_0+\delta) - \mathcal{L}(w_0) &= g^\top \delta \\ &\quad + \frac{1}{2}\sum_i q_i + \sum_{i<j} c_{ij} + \varepsilon_H(\delta), \\ q_i &= \|\delta_i\|_H^2, \\ c_{ij} &= \langle \delta_i, \delta_j\rangle_H. \end{aligned} \tag{3}$$

Here $\varepsilon_H$ contains both the mismatch between $H$ and $J_0$ and curvature variation along the path. If $H = J_0$ and the Hessian is $M$-Lipschitz in the relevant trust region, then $|\varepsilon_H(\delta)| \le M\|\delta\|_2^3/6$. Thus a small quadratic cost alone does not justify dropping the first-order term or evaluating only an interpolation point with $\epsilon < 1$.

**What near-orthogonality guarantees.** For $q_i q_j > 0$, define

$$\rho_H(i,j) = \frac{c_{ij}}{\sqrt{q_i q_j}}, \qquad \mu = \max_{i\neq j} |\rho_H(i,j)|. \tag{4}$$

For $k$ components in a common PSD metric,

$$\big(1-(k-1)\mu\big)\sum_i q_i \le \Big\|\sum_i \delta_i\Big\|_H^2 \le \big(1+(k-1)\mu\big)\sum_i q_i, \tag{5}$$

with a nontrivial lower bound when $(k-1)\mu < 1$. Consequently, near-zero $\rho_H(S, L)$ says that the sparse and low-rank repair costs are approximately additive. It does *not* say that their self costs are small, that either is orthogonal to quantization, or that the combination beats a single codec. Likewise, $\rho_H < 0$ is cancellation, not orthogonality. Achieving prescribed orthogonality further requires the repair degrees of freedom to span the relevant $H$-moments and to remain feasible after coefficient rounding and serialization (Appendix A).

**Comfort zones at a common rate.** Let $s_i \ge 0$ denote compression burden assigned to mechanism $i$ in a continuous relaxation, and let $d_i(s_i)$ be its local loss cost after its fixed codec charges have been paid. If the $d_i$ are differentiable and convex inside their comfort zones, an interior allocation solving min $\sum_i d_i(s_i)$ subject to $\sum_i s_i = S$ satisfies

$$d_i'(s_i) = \lambda \quad (s_i > 0), \qquad d_i'(0^+) \ge \lambda \quad (s_i = 0). \tag{6}$$

This separable condition assumes that allocation-dependent interactions are negligible or absorbed into the $d_i$; otherwise their partial derivatives enter the KKT condition. In particular, if a single mechanism $A$ carries all burden $S$ and $d_A'(S) > d_B'(0^+)$, shifting a sufficiently small amount to $B$ reduces the relaxed cost. This formalizes the comfort-zone intuition, but ranks only local candidates: ranks, sparse groups, descriptors, and alignment make the deployable problem discrete.

For a strict byte test, let $A$ be a realized reference endpoint at budget $B$, $u$ the degradation needed to free bytes, and $r$ a repair. A frontier claim additionally requires $A$ to be the best eligible single-mechanism endpoint under the same protocol. For the realized candidate $C = A + u + r$, define

$$\begin{aligned} P_A(u) &= \mathcal{L}(A+u) - \mathcal{L}(A), \\ G_A(r) &= \mathcal{L}(A) - \mathcal{L}(A+r), \\ \Gamma_A(u,r) &= \mathcal{L}(A+u+r) - \mathcal{L}(A+u) \\ &\quad - \mathcal{L}(A+r) + \mathcal{L}(A). \end{aligned} \tag{7}$$

Then the combination is better at the same rate whenever its actual serialized length obeys $\mathcal{B}(C) \le \mathcal{B}(A)$ and

$$G_A(r) > P_A(u) + \Gamma_A(u,r). \tag{8}$$

Locally, $\Gamma_A(u,r)$ contains the curvature cross term $u^\top H_A r$ plus proxy and path errors, where $H_A$ is the declared PSD proxy at $A$. Equation (8), not pairwise orthogonality, is the relevant sufficient condition.

**Repair efficiency must charge the file.** For a positive marginal file cost $m_r = \mathcal{B}(A+r) - \mathcal{B}(A)$, we report $\eta_{\text{NLL}} = G_A(r)/m_r$. The byte function includes codes, scales, sparse values, indices and row pointers, low-rank factors, descriptors, manifests, headers, and alignment/padding. Parameter count or value-only bits cannot substitute for this ledger. If OBS value re-estimation or a folded scale changes no final bytes, its direct held-out recovery is reported separately rather than as infinite efficiency; the zero-byte claim also requires decoding to contain no hidden state. A local quadratic ranking remains diagnostic until the final decoded endpoint is evaluated on held-out data.

These distinctions first appeared in one seed on six selected Pythia-70M MLP tensors: $S$ and $L$ are nearly orthogonal in the PSD proxy, both correlate negatively with $Q$, and a conservative QSL candidate is nevertheless worse than Q+L after tail padding to the same final bytes. Three additional, unpooled scalability-smoke jobs execute the same diagnostic over full Pythia MLP scope and depth-stratified OPT and Qwen MLP scopes. They repeat the near-orthogonal S–L category and positive strict-QSL-minus-QL endpoint sign for similarly underfilled candidates (Table 4), but they are still single-seed observations with unequal scopes. Multi-seed, common-scope, multi-rate, and production-format tests remain confirmatory requirements.

## 5. Experimental Protocol

**Verified exploratory endpoint.** The currently verified artifact uses the pinned Pythia-70M checkpoint (Biderman et al., 2023), WikiText-2 raw validation text (Merity et al., 2017), and six MLP projections from the first, middle, and last transformer blocks. Calibration and evaluation source texts are content-disjoint; fallback text is disabled. Held-out metrics cover 2,032 next-token predictions and dense perplexity 71.3525. The target selected-tensor artifact ratio is 0.258 relative to an FP16 reference. Six strategies are probed on 13-point paths, and 16 fixed contiguous windows provide paired descriptive differences. These windows are not treated as independent population samples.

**Completed scalability smoke.** We execute every native endpoint once at seed 17 and target selected-weight artifact ratio 0.258 on three scopes. Pythia-70M uses all 12 transformer-block MLP projections; OPT-125M (Zhang et al., 2022) uses `fc1/fc2` in five depth-stratified blocks (10 tensors); and Qwen3-0.6B (Yang et al., 2025) uses `up/down` projections in five blocks, excluding its gate projection (10 tensors). Every job evaluates 1,016 next-token predictions in eight fixed windows and probes six directions on the same 13-point path. All three share numerical source `d5444b26c3f7`, record a per-layer PSD covariance audit, and pass round-trip SHA and exact-file-byte checks. The jobs are three separate observations: each uses WikiText-2 raw validation text with content-disjoint calibration/evaluation source windows and fallback disabled; tokenizers and tensor scopes differ, and no cross-model metric is pooled or ranked.

**Preregistered and planned scale matrix.** The released 69-job configuration also declares Pythia-160M/410M and Llama-3.2-1B (Grattafiori et al., 2024), multiple rates, and additional seeds. Those rows remain unexecuted. Moreover, the present runner's seed argument does not alter its sequential data windows, so repeated invocations cannot be treated as independent confirmation. The separate eight-seed content-disjoint split manifest is preregistered but not yet consumed by this runner.

**Baselines and fairness.** Native comparisons include Q, global and block scales, Q+S, Q+S with OBS, Q+L, strict and relaxed QSL, Q+$\text{S}_{\text{OBS}}$+L, and component-scaled QSL. External methods enter a numeric table only if model checkpoint, W-only scope, tensor manifest, calibration/evaluation data, and exported physical payload match. We report data-free A0, calibrated no-backward A1, local backward-assisted B, and global fine-tuned/QAT C methods separately (Tables 1 and 5). Literature numbers never populate measured cells.

**Metrics.** Primary metrics are final file bytes, held-out NLL and perplexity change, normalized PSD proxy cost, and all pairwise signed $\rho_H$. For a positive byte increment we use held-out NLL recovery per added byte. Equal-byte re-estimation is reported as direct recovery, not infinite efficiency. Natural bytes, alignment, encoder wall time, peak memory, and provenance are secondary diagnostics.

## 6. Results and Analysis

### 6.1. Physical accounting exposes the strict disadvantage

Table 2 gives selected bounded endpoints; the complete native method set is retained in the released CSV. Q+L and strict component-scaled QSL are both 3,248,832 bytes, but strict component-scaled QSL is +0.558 perplexity worse. Its natural file is 3,233,152 bytes and receives 15,680 bytes of tail padding to match the 3,248,832-byte Q+L file. The control is therefore a feasible but under-filled, per-layer-guarded candidate rather than a budget-exhausted global frontier. Equal padded length prevents an overrun; it does not prove optimal use of the available natural bytes. The relaxed QSL endpoint adds only 5,056 bytes and is -0.286 perplexity relative to Q+L, but its paired interval crosses zero. This is evidence of a discrete threshold, not a confirmed frontier improvement.

### 6.2. Which small parameters are used effectively?

OBS is the clearest parameter-utilization result (Table 3 and Figure 1): it recovers 0.0269 mean NLL on an unchanged support/value payload and improves 16/16 windows. It therefore spends computation to make already-stored sparse values satisfy a local stationarity condition. Block scales recover 0.0744 NLL on all windows, but require 82,944 additional bytes. Q+L recovers more NLL per added byte than block scales in this tensor shape; the measured low-rank direction fits this residual more effectively, but one run does not establish a causal architectural advantage. The strict QSL exchange instead removes useful rank/support capacity to pay discrete sparse metadata and the legacy per-layer alignment guard, and wins only 4/16 windows against Q+L.

This suggests an analogue of scale protection for pruning: freeze a small support, then optimize its values (OBS) rather than buying more survivors. More generally, a repair is attractive when it changes existing stored state, spans a high-energy residual direction, and avoids a new descriptor or index threshold.

### 6.3. Near-orthogonality is not sufficient

At the strict endpoint, $\rho_H(S, L) = +0.030$, but $\rho_H(Q, S) = -0.423$ and $\rho_H(Q, L) = -0.572$ (Figure 4). S and L are nearly orthogonal to each other; both cancel part of Q. The three-way perturbation is not mutually orthogonal, and the small S–L cross term cannot compensate for their self costs or the capacity sacrificed to fit strict bytes. This is the observed counterpart of Equation (8).

The activation-Gram proxy tracks these particular paths closely (Figure 2), but every fitted linear coefficient is negative. Consequently, comparing only $\delta^\top H\delta$ omits a measurable first-order contribution. High correlation supports radial quadratic shape only along each already selected direction; it does not validate cross-candidate ranking and is not a certificate outside those directions or across models.

### 6.4. Three scalability-smoke observations

The strict scaled composite is higher-NLL than Q+L in all three decoded, final-byte-equal but naturally underfilled endpoints: by +0.0054 on Pythia, +0.0009 on OPT, and +0.0073 on Qwen. The fixed-window descriptive interval crosses zero for Pythia and OPT; Qwen favors Q+L in seven of eight windows and its interval remains positive. At the same time, $\rho_H(S, L)$ is +0.026, +0.087, and +0.022, respectively, whereas every Q–S and Q–L correlation is negative (Table 4 and Figure 3). Thus the coherence diagnostic remains compatible with local S–L additivity, but the realized exchange condition in Equation (8) fails for each tested conservative endpoint. Because unused natural bytes were not reallocated across layers, these observations do not reject every budget-exhausted composite.

The parameter-utilization controls are more mechanism-specific. Block scales improve Q in every job while paying positive descriptor/scale bytes, and OBS value re-estimation improves Q+S in all three without changing the sparse file. Folded global-scale re-estimation helps Pythia and OPT but hurts Qwen at the same bytes. This is precisely why zero-byte state reuse is attractive yet still requires held-out endpoint validation: low parameter cost does not imply a universally beneficial update. The committed CSVs contain all 33 endpoints and 234 path points. The appendix tabulates every endpoint and visualizes the 78 Q+L/strict path points in Table 6 and Figure 5; none is collapsed into a cross-model mean.

### 6.5. What would establish a combination advantage?

None of the current tail-padded strict candidates does relative to Q+L; OPT also contains a lower-NLL, slightly smaller Q+block-scale endpoint. A convincing positive result at the same compression rate must (i) use an identical tensor manifest and actual bytes, (ii) keep each component's self cost in its low-curvature regime, (iii) control all signed cross terms without calling cancellation orthogonality, (iv) clear the discrete rank/support/descriptor threshold, and (v) retain a positive held-out endpoint gain after the byte exchange. In the notation of Equation (8), these requirements reduce to an executable $G_A(r) > P_A(u) + \Gamma_A(u, r)$ test. The new global allocator and large-model method-ablation matrix are designed to test that inequality while allowing Q,

*Table 2.* Verified selected-tensor endpoints at target ratio 0.258. Scope: Pythia-70M, six MLP linear tensors, one seed, and 2,032 held-out tokens. "Natural" precedes final tail padding; "Final" is the complete research artifact whose stream offsets use 64-byte alignment. Lower $\Delta$NLL/$\Delta$PPL is better.

| Endpoint | Natural B | Final B | $\Delta$NLL | $\Delta$PPL | norm. $H$ |
|---|---|---|---|---|---|
| Q | 3,167,552 | 3,167,552 | +0.1982 | +15.644 | 0.01008 |
| Q + block scale | 3,250,496 | 3,250,496 | +0.1238 | +9.404 | 0.00577 |
| Q+S | 3,260,546 | 3,260,546 | +0.1224 | +9.293 | 0.00680 |
| Q+S (OBS) | 3,260,546 | 3,260,546 | +0.0956 | +7.156 | 0.00586 |
| **Q+L** | **3,248,832** | **3,248,832** | **+0.0905** | **+6.762** | **0.00463** |
| Q+S+L, strict | 3,233,152 | 3,248,832 | +0.0977 | +7.320 | 0.00507 |
| Q+S+L, +5,056 B | 3,253,888 | 3,253,888 | +0.0869 | +6.476 | 0.00457 |

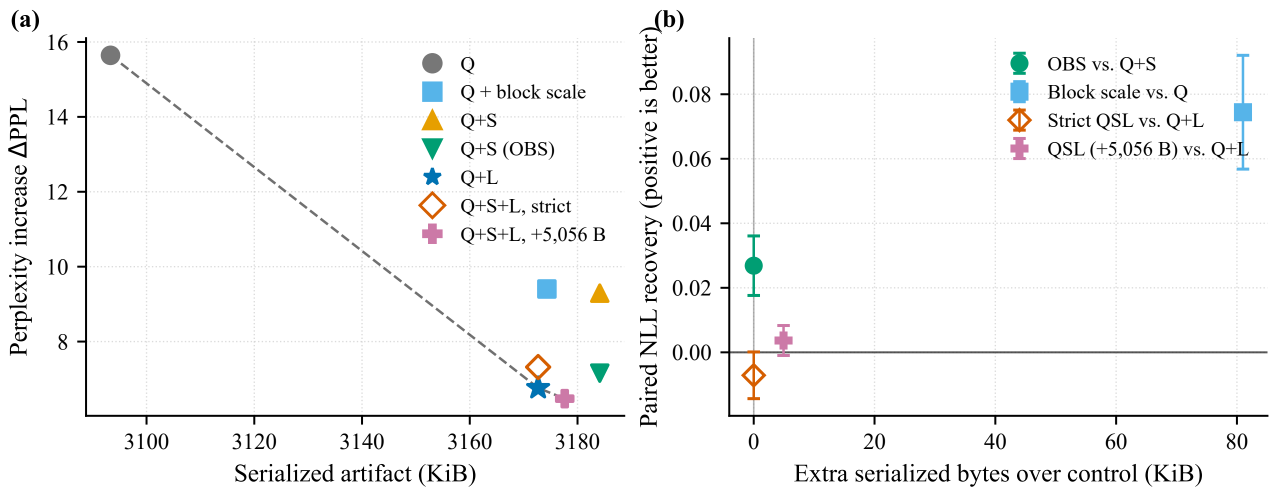


*Figure 1.* Verified exact-rate trade-offs from committed CSV artifacts. (a) Physical bytes versus perplexity increase; only measured artifact files are plotted. (b) Paired held-out NLL recovery and byte overhead for targeted repairs. Error bars are descriptive normal intervals over 16 fixed windows, not population confidence intervals. Scope is one seed and six selected Pythia-70M MLP tensors.

*Table 3.* Paired repair effects in the same bounded run. Positive recovery favors the left method. Intervals are descriptive over fixed windows.

| Comparison | $\Delta$ bytes | NLL recovery | 95% interval | Wins |
|---|---|---|---|---|
| OBS vs. Q+S | +0 | +0.0269 | [+0.0176, +0.0361] | 16/16 |
| Block scale vs. Q | +82,944 | +0.0744 | [+0.0568, +0.0921] | 16/16 |
| Strict QSL vs. Q+L | +0 | -0.0071 | [-0.0144, +0.0001] | 4/16 |
| QSL (+5,056 B) vs. Q+L | +5,056 | +0.0037 | [-0.0010, +0.0083] | 10/16 |

Q+$S_{OBS}$, and Q+L as nested degenerate states. Until those jobs complete, they remain planned and are not silently promoted to results.

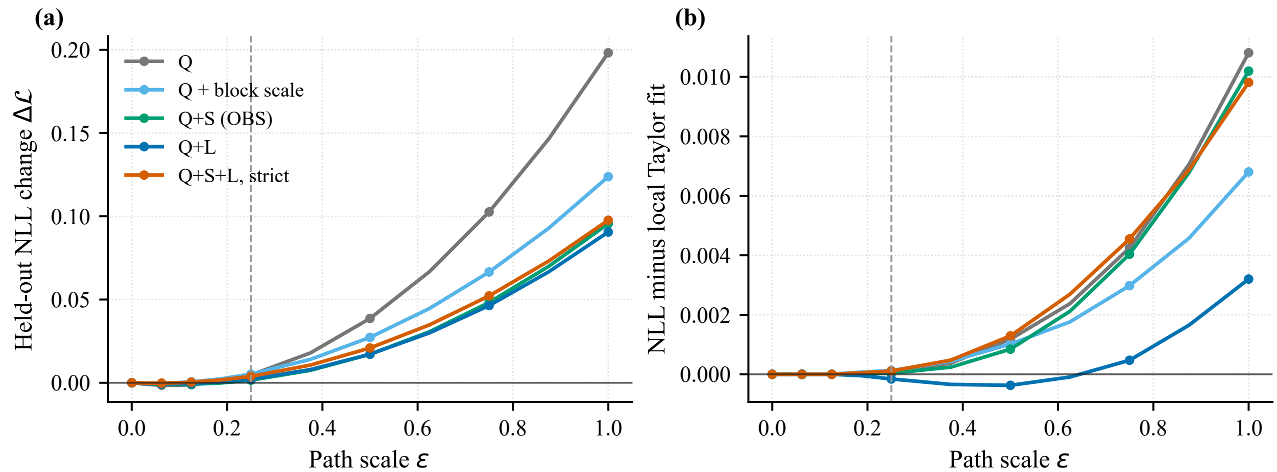


*Figure 2.* Held-out loss landscapes for six codec directions in the original bounded probe. Left: NLL along the 13-point interpolation. Right: residual after the local linear–quadratic fit. Proxy/path correlations range from 0.9970 to 0.9995, yet nonzero linear terms and endpoint residuals require final NLL evaluation. Scope is one seed, six selected Pythia-70M MLP tensors, and 2,032 held-out tokens.

*Table 4.* Single-seed scalability smoke at target selected-weight artifact ratio 0.258. Pythia covers all MLP projections, while OPT and Qwen use ten depth-stratified MLP projections. Bytes are complete aligned research-codec files for selected tensors, not whole-model or production payloads. "Strict−QL" is the signed held-out NLL difference after tail padding to identical final bytes; the strict natural files leave 31,360/640/1,024 bytes unused for Pythia/OPT/Qwen. Negative favors the composite. Rows are not pooled or ranked across models.

| Model / scope | Tensors / M params | QL=strict bytes (rate) | ΔNLL QL | ΔNLL strict | Strict−QL | $\rho_{SL}$ | $\rho_{QS}/\rho_{QL}$ |
|---|---|---|---|---|---|---|---|
| Pythia-70M / full MLP | 12 / 12.58 | 6,497,344 (0.2581) | +0.1963 | +0.2017 | +0.0054 | +0.026 | -0.334/-0.588 |
| OPT-125M / 5-depth MLP | 10 / 23.59 | 12,158,976 (0.2577) | +0.0735 | +0.0744 | +0.0009 | +0.087 | -0.367/-0.611 |
| Qwen3-0.6B / 5-depth MLP | 10 / 31.46 | 16,196,352 (0.2574) | +0.0226 | +0.0299 | +0.0073 | +0.022 | -0.521/-0.743 |

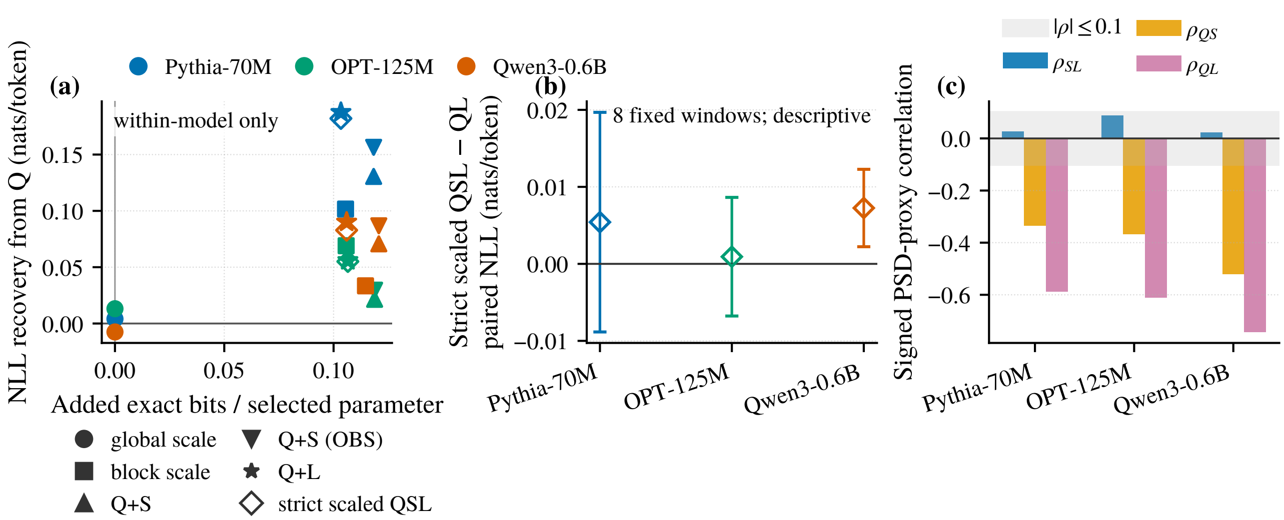


*Figure 3.* Mechanism behavior in three separate seed-17 scalability-smoke jobs. (a) Within each color/model, held-out NLL recovery from Q versus added exact file bits per selected parameter; the zero-byte scale point is direct folded re-estimation, not infinite efficiency. (b) Conservative strict component-scaled QSL minus Q+L after tail padding to identical final file bytes; the strict natural files underfill their Q+L caps by 31,360, 640, and 1,024 bytes for Pythia, OPT, and Qwen, respectively; error bars are descriptive mean ±1.96 sample standard errors over eight fixed windows, not confidence intervals or significance tests. (c) Signed activation-Gram correlations; the band marks $|\rho| \le 0.1$, while negative Q–S/Q–L values denote cancellation. Tokenizers and tensor scopes differ, so magnitudes are not ranked across models and rates are not whole-model compression ratios.

## 7. Limitations and Conclusion

The strongest detailed probe remains one seed on six selected Pythia-70M tensors, now accompanied by three separate seed-17 scalability-smoke jobs over full Pythia MLP and depth-stratified OPT/Qwen MLP scopes. Their repeated sign is not multi-seed confirmation: the activation Gram is a PSD reconstruction proxy rather than the task Hessian, fixed windows do not support population inference, and unequal scopes forbid a model-scale trend. The research codec also does not establish kernel speed, whole-checkpoint size, or energy savings. External methods remain literature-reported until their training budgets and physical payloads are reproduced under a matched protocol.

Within that boundary, the result is consistent across the executed scopes: sparse and low-rank repairs can be nearly orthogonal without a same-byte composite advantage. Exact serialization reveals the discrete cost that logical accounting hides. OBS improves the already-paid sparse payload in all three smoke jobs, whereas zero-byte global-scale re-estimation has mixed sign; parameter thrift therefore helps only when the fitted update also lowers the held-out endpoint. Composite compression is most plausible when mechanisms exchange real bytes at favorable held-out marginal cost. Orthogonality is one diagnostic in that decision, not the decision itself.

## Impact Statement

This work aims to make language-model compression comparisons more reproducible by charging actual serialized state and by separating diagnostic proxies from held-out endpoint evidence. Better compression can reduce storage and inference-resource requirements, but accuracy losses can be uneven across tasks or populations. The research artifact codec is not a production runtime, and the present experiments do not establish deployment speed, energy savings, safety preservation, or fairness preservation; those properties require separate evaluation before use.

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# A. Proofs and Scope of the Curvature Analysis

## A.1. Taylor expansion and the role of a PSD proxy

Let $\mathcal{L} : \mathbb{R}^p \to \mathbb{R}$ be twice continuously differentiable on the segment from $w_0$ to $w_0 + \delta$. With $J(w) = \nabla^2 \mathcal{L}(w)$, Taylor's integral form gives

$$\mathcal{L}(w_0+\delta) - \mathcal{L}(w_0) = g^\top \delta + \int_0^1 (1-t)\, \delta^\top J(w_0+t\delta)\delta \, dt. \tag{9}$$

Writing $J_0 = J(w_0)$ yields the usual quadratic term and a path remainder. If $\|J(w_0+t\delta) - J_0\|_{\mathrm{op}} \le Mt\|\delta\|_2$, then

$$\left|\mathcal{L}(w_0+\delta) - \mathcal{L}(w_0) - g^\top \delta - \tfrac{1}{2}\delta^\top J_0 \delta\right| \le \frac{M}{6}\|\delta\|_2^3, \tag{10}$$

because $\int_0^1 t(1-t)dt = 1/6$.

The task Hessian $J_0$ can be indefinite. For geometric diagnostics we instead select and disclose $H \succeq 0$. Combining proxy mismatch and path variation gives

$$\begin{aligned} \varepsilon_H(\delta) &= \tfrac{1}{2}\delta^\top (J_0 - H)\delta + R_3(\delta), \\ |\varepsilon_H(\delta)| &\le \tfrac{1}{2}\|J_0 - H\|_{\mathrm{op}}\|\delta\|_2^2 \\ &\quad + \tfrac{M}{6}\|\delta\|_2^3. \end{aligned} \tag{11}$$

The bound need not be numerically tight; its purpose is to expose the assumption hidden when a PSD surrogate is called "the Hessian." For a damped task Hessian, $H = J_0 + \lambda I$ is a metric only when $\lambda \ge -\lambda_{\min}(J_0)$. Damping must therefore be reported, and it changes the measured angles. If $H$ is singular, its inner product is understood on the quotient by $\ker(H)$; correlations with $q_i = 0$ are undefined rather than zero.

For sequential maps $T_1, \dots, T_k$, let $w_i = T_i(w_{i-1})$ and $\delta_i = w_i - w_{i-1}$. Then $w_k - w_0 = \sum_i \delta_i$, so substituting into the $H$-quadratic gives

$$\frac{1}{2}\Big\|\sum_i \delta_i\Big\|_H^2 = \frac{1}{2}\sum_i \|\delta_i\|_H^2 + \sum_{i<j}\langle \delta_i, \delta_j\rangle_H. \tag{12}$$

This algebra does not make the components order invariant. If $T_iT_j(w) \neq T_jT_i(w)$, changing order changes both increments and the final endpoint; all terms must be recomputed after final decoding.

## A.2. Coherence bound

**Proposition 1 (PSD coherence).** Let $H \succeq 0$, let $q_i = \|\delta_i\|_H^2 > 0$, and suppose $|\rho_H(i,j)| \le \mu$ for every $i \neq j$. Then Equation (5) holds.

**Proof.** Cauchy–Schwarz in the PSD seminorm gives $|\langle \delta_i, \delta_j \rangle_H| \le \mu\sqrt{q_i q_j}$. Moreover,

$$2\sum_{i<j}\sqrt{q_i q_j} = \left(\sum_i \sqrt{q_i}\right)^2 - \sum_i q_i \le (k-1)\sum_i q_i, \tag{13}$$

where the last inequality follows from $(\sum_i \sqrt{q_i})^2 \le k\sum_i q_i$. Applying the upper and lower signs to the cross-term expansion proves the claim. □

The proposition controls only additivity of the PSD quadratic. It does not control $g^\top \delta$, proxy error, path remainder, component self costs, or serialized rate. A negative correlation may reduce the quadratic through cancellation, but $|\rho_H|$ is then large and the components are not near-orthogonal.

## A.3. Feasibility of enforced orthogonality

Suppose an error $e$ can be modified using columns $B = [b_1, \dots, b_d]$ and we want the result to be $H$-orthogonal to columns $U = [u_1, \dots, u_m]$. A coefficient vector $\alpha$ must solve

$$A\alpha = -c, \qquad A = U^\top HB, \qquad c = U^\top He. \tag{14}$$

An unconstrained solution exists if and only if $-c$ lies in the range of $A$. Thus a single global scale generally cannot remove several independent moments. Codec feasibility adds stronger conditions: $\alpha$ must respect the allowed scale range, support or rank, storage precision, and descriptor layout. After rounding, the residual $\|A\widehat{\alpha} + c\|$ must be remeasured. Subsequent requantization, support selection, or rank truncation can reintroduce the moments, so orthogonality is a property of the final decoded endpoint, not of an intermediate floating-point solve.

## A.4. A sufficient endpoint comparison

Define the local PSD score

$$\widehat{\Delta}_H(z) = g^\top z + \tfrac{1}{2} z^\top H z \tag{15}$$

and suppose a candidate-specific certificate satisfies

$$\left|\mathcal{L}(w_0+z) - \mathcal{L}(w_0) - \widehat{\Delta}_H(z)\right| \le r(z). \tag{16}$$

The radius $r(z)$ must include proxy mismatch and path curvature; it may be estimated conservatively from held-out interpolation diagnostics but is not supplied by coherence alone.

**Proposition 2 (same-byte sufficient condition).** Let $z_C$ be a composite candidate and $\{z_j\}$ the eligible single-mechanism candidates. If

$$\begin{aligned} &\mathcal{B}(z_C) \le B, \\ &\widehat{\Delta}_H(z_C) + r(z_C) < \min_{j:\,\mathcal{B}(z_j)\le B}\left\{\widehat{\Delta}_H(z_j) - r(z_j)\right\}. \end{aligned} \tag{17}$$

then the composite has lower loss than every eligible single candidate at budget $B$.

**Proof.** The left side of the strict inequality upper-bounds the composite loss increment, while every term on the right lower-bounds the corresponding single-candidate increment. $\square$

This proposition is deliberately sufficient, not necessary. In the absence of a trustworthy $r(z)$, held-out endpoint loss replaces the certificate; a high proxy/path correlation is a diagnostic, not a proof of the inequality.

### A.5. Comfort-zone allocation

Let each $d_i : [0, S] \to \mathbb{R}$ be differentiable, convex, and nondecreasing in assigned compression burden. Consider

$$\min_{s_i \geq 0} \sum_{i=1}^{k} d_i(s_i) \quad \text{subject to} \quad \sum_{i=1}^{k} s_i = S. \tag{18}$$

**Proposition 3 (marginal-cost allocation).** At an optimum of Equation (18), every active component satisfies $d_i'(s_i) = \lambda$ and every inactive component satisfies $d_i'(0^+) \geq \lambda$. If the all-$A$ allocation has $d_A'(S) > d_B'(0^+)$ and both derivatives are continuous near the endpoint, then an allocation using both $A$ and $B$ has strictly lower cost.

**Proof.** The KKT stationarity and complementary-slackness conditions are

$$d_i'(s_i) - \lambda - \nu_i = 0, \qquad \nu_i \geq 0, \qquad \nu_i s_i = 0.$$

They give the stated active and inactive conditions. For the second claim, set $f(\epsilon) = d_A(S - \epsilon) + d_B(\epsilon)$. Then $f'(0^+) = -d_A'(S) + d_B'(0^+) < 0$, so sufficiently small positive $\epsilon$ improves the objective. $\square$

Separability is an assumption of Equation (18). For an objective $\sum_i d_i(s_i) + \gamma(s)$ with non-negligible allocation-dependent interactions, the active condition becomes $d_i'(s_i) + \partial_i \gamma(s) = \lambda$.

The proposition explains why several mechanisms can outperform one mechanism pushed beyond its comfortable region, but only in the stated continuous model. A codec has a lattice of feasible endpoints: one rank, one sparse group, or a new descriptor may have a fixed charge. The deployable analogue is a finite exact-byte exchange.

Let $A$ be a realized endpoint, $u$ a perturbation that frees payload, and $r$ a repair perturbation. For a fixed, order-specific decomposition $C = A + u + r$, direct addition and subtraction gives

$$\mathcal{L}(C) - \mathcal{L}(A) = P_A(u) - G_A(r) + \Gamma_A(u, r). \tag{19}$$

Hence Equation (8) is equivalent to a strict improvement once $\mathcal{B}(C) \leq \mathcal{B}(A)$. Around $A$, the second-order part of $\Gamma_A$ is $u^\top J_A r$; replacing $J_A$ by a PSD proxy again adds a mismatch and path term. This identity also shows why repair-to-repair orthogonality is insufficient: the cost of freeing bytes and all interactions still remain.

### A.6. Physical bytes and parameter utilization

For endpoint $z$, the comparison rate is the on-disk length

$$\begin{aligned}\mathcal{B}(z) = B_{\text{header}} &+ B_{\text{manifest}} + B_{\text{descriptor}} \\ &+ B_{\text{codes}} + B_{\text{scales}} \\ &+ B_{\text{sparse values}} + B_{\text{indices}} \\ &+ B_{\text{rowptr}} + B_{\text{factors}} + B_{\text{alignment}} \\ &+ B_{\text{tail padding}}.\end{aligned} \tag{20}$$

The marginal cost $m_r = \mathcal{B}(A + r) - \mathcal{B}(A)$ is therefore integer-valued and need not equal the sum of value streams. Alignment can make it nonadditive. We disclose both natural bytes and final file bytes: the latter decides the strict budget, while the former reveals when a nominally zero-cost change merely consumes padding slack.

When $m_r > 0$, held-out recovery per byte is

$$\eta_{\text{NLL}}(r \mid A) = \frac{\mathcal{L}(A) - \mathcal{L}(A + r)}{m_r}. \tag{21}$$

Suppose $u$ frees $f_u > 0$ bytes and has loss penalty $P_A(u) = f_u \eta_{\text{giveup}}$. If the repair has $G_A(r) = m_r \eta_{\text{repair}}$, then a feasible exchange improves loss when

$$m_r \eta_{\text{repair}} > f_u \eta_{\text{giveup}} + \Gamma_A(u, r), \qquad m_r \leq f_u, \tag{22}$$

with any remaining slack represented in the actual serialization. For an exactly matched exchange $m_r = f_u$, this reduces to an efficiency comparison corrected by interaction per byte. Merely comparing the two ratios is not sufficient when the exchanged byte amounts differ.

If $m_r = 0$, division would create an artificial infinite efficiency. Instead we report the direct recovery, verify that the decoded artifact contains no side state, and state whether natural bytes or only final file bytes are unchanged. Fixed-support OBS can satisfy this condition when re-estimated values replace already stored values; a scale can do so only if it is demonstrably folded into an already stored field.

### A.7. Limits of the diagnostic

The analysis is local and order specific. Its strongest threats are: (i) noncommuting quantization, support selection, and low-rank fitting; (ii) curvature drift between $w_0$ and the decoded endpoint; (iii) an activation/Fisher/GGN proxy that does not preserve task-loss ranking; (iv) calibration geometry that does not transfer to held-out text, seeds, layers, or model families; and (v) discrete metadata and alignment thresholds. These threats motivate final-endpoint NLL, multiple independent splits, source and artifact hashes, and full byte round trips. The strongest fine-grained numerical evidence remains one seed and six selected MLP linear tensors of Pythia-70M. Three additional, unpooled seed-17 scalability-smoke jobs cover full Pythia-70M MLP scope and depth-stratified OPT-125M and Qwen3-0.6B MLP scopes at one selected-weight artifact rate. The propositions and smoke observations do not upgrade these probes into a multi-seed, whole-model, or cross-model advantage claim.

## B. Reproducibility and Evidence Status

**Four evidence labels.** Repository claims are tagged `verified`, `preregistered`, `planned`, or `literature-reported`. Verified numeric claims must resolve to committed endpoint CSVs and artifact manifests. Preregistered indicates a fixed protocol without completed model outputs. Planned rows are design commitments, not results. Literature-reported rows are protocol metadata from primary papers or official repositories and are never merged with our measured table.

**Verified run.** The bounded run uses Pythia-70M revision `a39f36b100fe` (the full hash is in the run manifest). It selects six MLP linear tensors, evaluates 2,032 held-out next tokens, emits all strategy artifacts, and stores natural/file bytes, SHA-256, round-trip status, window NLL, 13-point paths, source/environment metadata, and process completion evidence. The numerical run predates source-file SHA capture for its then-dirty working tree; this provenance limitation is disclosed rather than reconstructed after the fact.

**Scale suite.** `scripts/run_large_scale_hessian_suite.py` expands a declarative model/seed/rate/scope matrix. It injects the repository's `src` directory into the numerical subprocess, records config/job/source fingerprints, and validates the exact selected tensor count and all artifact hashes. `RUNNING`, `FAILED`, stale partial directories, and source drift are fail-closed. Optional models require explicit selection. The completed pilot has three `completed_valid` jobs, 33 endpoint files, 12 fixed-window comparisons, and 234 path rows. Their numerical source is `d5444b26c3f7`. Each selected layer records the original minimum eigenvalue relative to spectral scale, the float32 PSD floor, applied diagonal shift, and final minimum eigenvalue; every downstream metric consumes the same validated covariance. The completed 2026-07-14 pilot used a minimum-eigenvalue reduction with a zero initializer, so its positive minima were conservatively reported as zero; those artifacts close the shift ledger and final nonnegativity but do not claim an observed strictly positive stored floor. The revised, unexecuted runner reports the true minimum and separately records the whitened-SVD fitting floor. Both establish a usable PSD proxy, not equality with the task Hessian.

**Paper regeneration.** Tables and numeric macros are generated by `paper/results/build_verified_tables.py` and the fail-closed scale aggregate by `paper/results/build_scaling_pilot.py`. The latter verifies suite/job/source fingerprints, checkpoint revisions, every physical file SHA, Hessian decomposition, fixed windows, and Taylor paths before producing a table or figure input. Each plot script in `paper/figures` reads committed CSVs and writes PDF, SVG, PNG, and a manifest with input hashes. No experimental point in this manuscript is hand-entered into a plotting script. The paper uses the official ICML 2026 style and can be compiled with Tectonic 0.16.9 using the commands documented in `paper/README.md`.

**External-method audit.** The selected paper matrix records objective, updated state, calibration/backward requirement, tensor and W/A/KV scope, payload fields, official code status, and direct-comparison eligibility. In particular, LiftQuant block correction is a backward-assisted lane, while its optional end-to-end correction is a global training lane. A compatibility-patched layer-0 smoke validates only one control path and artifact write; it supplies no full-model PPL or same-byte result. The repository retains a broader 59-row method registry. Ten D-lane rows separately record multimodal/video quantization, cache, sparse or structured attention, and sparse–low-rank attention; they are scope-mismatched mechanism references rather than direct LLM weight-PTQ controls. The 27-row table below remains the primary-source subset most directly tied to this paper's weight mechanisms and comparison lanes.

**Multimodal supplement.** Video-DiT quantizers use temporal or attention-guided distillation (Feng et al., 2025b;a); QuantSparse jointly optimizes quantization and sparse attention (Feng et al., 2026); and CacheQuant jointly schedules diffusion caching and quantization (Liu et al., 2025b). TeaCache, Sparse VideoGen, and Sparse-vDiT cover timestep reuse and spatial/temporal sparse patterns (Liu et al., 2025a; Xi et al., 2025; Chen et al., 2025), while VMonarch, MonarchRT, and RoPeSLR retain structured, dense-mixing, or low-rank attention components (Liang et al., 2026; Agarwal et al., 2026; Liu et al., 2026). The repository supplement proposes a held-out gate among sink/global low rank, local structure, dynamic sparse routes, and dense fallback, jointly accounted with W/A PTQ, cache, and temporal reuse. This is planned work: paper-reported speedups are scope-specific and are never multiplied into a claimed stack.

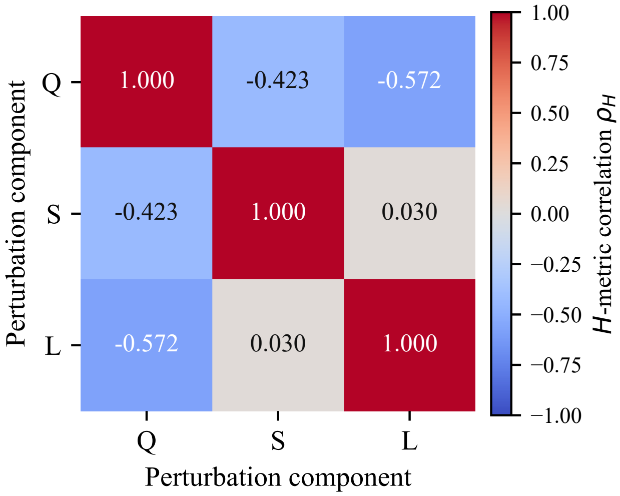


*Figure 4.* Signed activation-Gram correlations for the original conservative QSL endpoint, whose natural encoding is tail-padded to the Q+L final byte count. S–L is near zero, while Q–S and Q–L are negative cancellation. Scope is one seed and six selected Pythia-70M MLP tensors.

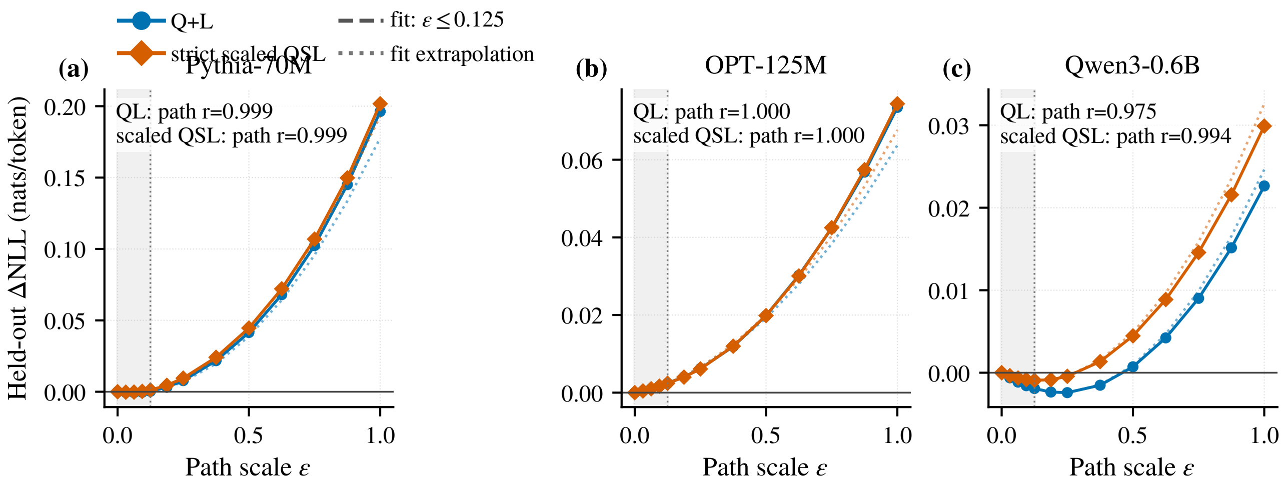


*Figure 5.* Thirteen-point one-dimensional interpolation probes for Q+L and strict component-scaled QSL in the three separate scalability-smoke jobs. Solid points are measured held-out NLL changes. Dashed curves are fitted only on $\epsilon \leq 0.125$; lighter dotted continuations are extrapolations. The shaded region marks the fit interval, and each annotation is the path-specific proxy–NLL correlation over 13 points. Panels use independent NLL scales and different tokenizers, so their magnitudes are not compared. These slices are not a global loss landscape or multi-seed evidence.

*Table 5.* Primary-source comparison audit. Eligibility is determined by training dependence and exported state, not nominal bits alone. Direct comparison additionally requires a matched checkpoint, tensor scope, data, and serializer. Every external row is literature-reported.

| Method | Venue | Lane | Encoder process | Deployed state to charge | Exact file bytes? |
|---|---|---|---|---|---|
| GPTQ (Frantar et al., 2023) | ICLR'23 | A1 | Activation Hessian proxy; no backward | Codes, scales, grouping/order metadata | No† |
| SparseGPT (Frantar & Alistarh, 2023) | ICML'23 | A1 | Second-order one-shot pruning | Values and explicit support/N:M layout | No† |
| Wanda (Sun et al., 2024) | ICLR'24 | A1 | Weight–activation magnitude; no update | Values and explicit support/N:M layout | No† |
| AWQ (Lin et al., 2024) | MLSys'24 | A1 | Activation-aware scale/clipping search | Codes, scales and non-folded transforms | No† |
| SpQR (Dettmers et al., 2024) | ICLR'24 | A1 | Second-order outlier separation | Bulk codes plus outlier values/indices | No† |
| LQER (Zhang et al., 2024) | ICML'24 | A1 | Activation-scaled closed-form SVD | Base codec plus low-rank factors | No† |
| QERA (Zhang et al., 2025) | ICLR'25 | A1 | Closed-form reconstruction (PTQ row) | Base codec plus low-rank factors | No† |
| EoRA (Liu et al., 2024) | arXiv'24 | A1 | Eigenspace low-rank approximation | Compressed base plus low-rank factors | No† |
| SLiM (Mozaffari et al., 2025) | ICML'25 | A1 | One-shot quantization, 2:4 pruning, LR repair | Codes, 2:4 layout and low-rank factors | No† |
| Harma et al. (Harma et al., 2025) | ICLR'25 | A0 | Q/S order and non-orthogonality analysis | Sparse quantized values, scales and support | No† |
| OBR (Guo et al., 2026) | ICLR'26 | A1 | Closed-form Hessian Q/S compensation | Folded sparse low-bit weights and layout | No† |
| FOEM (Zheng et al., 2026) | AAAI'26 | A1 | First-order proxy without backpropagation | Base-codec state after compensation | No† |
| Radio (Young, 2025) | ICML'25 | A1 | Post-training rate–distortion allocation | Codes plus allocation/decoder metadata | No† |
| Q-Palette (Lee & Song, 2025) | NeurIPS'25 | A0/A1 | Data-free or calibrated mixed-scheme search | Codes, codebooks, scheme/fusion metadata | No† |
| SRR (Cho et al., 2026) | ICML'26 | A1 | Closed-form preserve/reconstruct rank split | Base codec plus both low-rank parts | No† |
| SERQ (Park et al., 2026) | ICLR'26 | A1 | Offline saliency decomposition/permutation | W/A codec, one LR matrix, permutation | No† |
| ResComp (Li et al., 2026) | ICLR'26 | A1 | Sequential compensation; no global tuning | Base codec plus any unabsorbed state | No† |
| SqueezeLLM (Kim et al., 2024) | ICML'24 | B | Loss-gradient-square/Fisher checkpoint | Indices, codebooks and sparse outliers | No† |
| GuidedQuant (Kim et al., 2025) | ICML'25 | B | End-loss backward/Fisher blocks | Attached base-quantizer payload | No† |
| Hessian ECSQ (Choi et al., 2017) | ICLR'17 | B | Hessian-weighted entropy-constrained coding | Code stream, codebook/LUT and rate metadata | No† |
| Optimal Formats (Orr et al., 2025) | arXiv'25 | B* | Variable-length formats; Fisher allocation | Streams, code tables and format allocation | No† |
| SliderQuant (Wang et al., 2026) | ICLR'26 | B | Learned sliding-window scales and rank-4 repair | Codes plus any unmerged scale/low-rank state | No† |
| YAQA (Tseng et al., 2026) | ICML'26 | B | Full-model KL HVP/backward sketches | Quantizer payload; sketches are encoder-only | No† |
| LiftQuant (block) (He et al., 2026) | ICML'26 | B | Block-output correction with backward passes | Lifted codes and projection/correction state | No† |
| LiftQuant (E2E) (He et al., 2026) | ICML'26 | C | Global causal-LM recovery | Learned lifted/projection/correction state | No† |
| FPTQuant (van Breugel et al., 2026) | ICML'26‡ | C | Locally and end-to-end learned transforms | Transformed codec and dynamic transform state | No† |
| ProjQ (Yu et al., 2026) | ICML'26 | C | Orthogonal Q/L projection plus adaptation | Base codec, adapter and projection state | No† |

A0: data-free frozen compression; A1: calibrated/no-backward frozen compression; B: encoder-side backward/HVP without global recovery; C: learned or fine-tuned recovery. Optional QPEFT/PEFT results in QERA, SLiM, YAQA, and SRR are excluded from their A/B rows. *Optimal Formats also contains a data-free format-analysis lane; B refers only to Fisher allocation. †The primary source reports bit width, sparsity, rank, logical memory, or runtime memory, but this audit found no complete deployable output-file byte enumeration including all codebooks, supports, factors, scales, headers, padding, and alignment. ‡The official ICML 2026 listing and current arXiv metadata identify ICML 2026; an older OpenReview listing identifies ICLR 2026.

*Table 6.* All native endpoints for the three scalability-smoke jobs. Rate and bits/parameter charge complete selected-tensor artifact files. The scopes and tokenizers differ, so rows must only be compared within each model block.

| Model | Endpoint | Bytes | Rate | bits/sel. param | ΔNLL | ΔPPL | norm. $H$ | Repair (nnz/rank/dof) |
|---|---|---|---|---|---|---|---|---|
| Pythia-70M | Q | 6,334,784 | 0.2517 | 4.028 | +0.3838 | +43.128 | 0.00961 | 0/0/0 |
| | Q + global scale | 6,334,784 | 0.2517 | 4.028 | +0.3797 | +42.574 | 0.00946 | 0/0/12 |
| | Q + block scale | 6,500,672 | 0.2583 | 4.133 | +0.2821 | +30.053 | 0.00597 | 0/0/98304 |
| | Q+S | 6,520,898 | 0.2591 | 4.146 | +0.2532 | +26.562 | 0.00704 | 27276/0/0 |
| | Q+S (OBS) | 6,520,898 | 0.2591 | 4.146 | +0.2273 | +23.533 | 0.00611 | 27276/0/27276 |
| | Q+L | 6,497,344 | 0.2581 | 4.131 | +0.1963 | +19.994 | 0.00476 | 0/30/0 |
| | Q+S+L strict | 6,497,344 | 0.2581 | 4.131 | +0.2060 | +21.094 | 0.00530 | 5248/16/0 |
| | Q+S+L strict + scale | 6,497,344 | 0.2581 | 4.131 | +0.2017 | +20.603 | 0.00526 | 5248/16/30 |
| | Q+S+L relaxed | 6,507,456 | 0.2585 | 4.137 | +0.2028 | +20.733 | 0.00475 | 6470/23/0 |
| | Q+S (OBS)+L | 6,507,456 | 0.2585 | 4.137 | +0.1988 | +20.276 | 0.00476 | 6470/23/6470 |
| | Q+S+L relaxed + scale | 6,507,456 | 0.2585 | 4.137 | +0.1993 | +20.333 | 0.00471 | 6470/23/30 |
| OPT-125M | Q | 11,844,800 | 0.2510 | 4.016 | +0.1295 | +7.744 | 0.00585 | 0/0/0 |
| | Q + global scale | 11,844,800 | 0.2510 | 4.016 | +0.1163 | +6.913 | 0.00576 | 0/0/10 |
| | Q + block scale | 12,156,672 | 0.2576 | 4.122 | +0.0604 | +3.491 | 0.00337 | 0/0/175104 |
| | Q+S | 12,195,064 | 0.2584 | 4.135 | +0.1080 | +6.390 | 0.00420 | 65560/0/0 |
| | Q+S (OBS) | 12,195,064 | 0.2584 | 4.135 | +0.0999 | +5.884 | 0.00334 | 65560/0/65560 |
| | Q+L | 12,158,976 | 0.2577 | 4.123 | +0.0735 | +4.272 | 0.00279 | 0/40/0 |
| | Q+S+L strict | 12,158,976 | 0.2577 | 4.123 | +0.0828 | +4.835 | 0.00309 | 10528/23/0 |
| | Q+S+L strict + scale | 12,158,976 | 0.2577 | 4.123 | +0.0744 | +4.328 | 0.00307 | 10528/23/30 |
| | Q+S+L relaxed | 12,201,984 | 0.2586 | 4.138 | +0.0808 | +4.716 | 0.00305 | 13720/27/0 |
| | Q+S (OBS)+L | 12,201,984 | 0.2586 | 4.138 | +0.0804 | +4.692 | 0.00302 | 13720/27/13720 |
| | Q+S+L relaxed + scale | 12,201,984 | 0.2586 | 4.138 | +0.0727 | +4.224 | 0.00303 | 13720/27/30 |
| Qwen3-0.6B | Q | 15,779,648 | 0.2508 | 4.013 | +0.1127 | +3.560 | 0.02295 | 0/0/0 |
| | Q + global scale | 15,779,648 | 0.2508 | 4.013 | +0.1202 | +3.812 | 0.02204 | 0/0/10 |
| | Q + block scale | 16,230,272 | 0.2580 | 4.128 | +0.0793 | +2.462 | 0.00662 | 0/0/245760 |
| | Q+S | 16,254,218 | 0.2583 | 4.134 | +0.0420 | +1.280 | 0.00669 | 95090/0/0 |
| | Q+S (OBS) | 16,254,218 | 0.2583 | 4.134 | +0.0260 | +0.786 | 0.00386 | 95090/0/95090 |
| | Q+L | 16,196,352 | 0.2574 | 4.119 | +0.0226 | +0.683 | 0.00414 | 0/50/0 |
| | Q+S+L strict | 16,196,352 | 0.2574 | 4.119 | +0.0303 | +0.918 | 0.00444 | 17440/30/0 |
| | Q+S+L strict + scale | 16,196,352 | 0.2574 | 4.119 | +0.0299 | +0.905 | 0.00440 | 17440/30/30 |
| | Q+S+L relaxed | 16,261,248 | 0.2584 | 4.135 | +0.0224 | +0.676 | 0.00415 | 25458/34/0 |
| | Q+S (OBS)+L | 16,261,248 | 0.2584 | 4.135 | +0.0224 | +0.675 | 0.00419 | 25458/34/25458 |
| | Q+S+L relaxed + scale | 16,261,248 | 0.2584 | 4.135 | +0.0219 | +0.660 | 0.00411 | 25458/34/30 |